

Exercises 10.B

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Exercise 1. Suppose V is a real vector space. Suppose $T \in \mathcal{L}(V)$ has no eigenvalues. Prove that $\det T > 0$.

Proof. Since T has no eigenvalues, T_C has no "real" eigenvalues. That is, each eigenvalue of T_C has a nontrivial real part. This implies that all the eigenvalues of T_C come in pairs of a complex number and its conjugate, i.e. if λ is an eigenvalue of T_C , then $\bar{\lambda}$ is also an eigenvalue of T_C . Then the determinant of T , which is the product of the eigenvalues of T_C , is of the form

$$(a_1 + b_1 i)(a_1 - b_1 i) \cdots (a_n + b_n i)(a_n - b_n i),$$

where $a_j, b_j \in \mathbb{R}$ for all $j \in \{1, \dots, n\}$. Simplifying this, we get

$$(a_1^2 + b_1^2) \cdots (a_n^2 + b_n^2),$$

which is a product of *positive* integers since a_j and b_j are not both 0 for a given $j \in \{1, \dots, n\}$ (otherwise 0 would be an eigenvalue of T). Thus the product, which is the determinant, is positive. $\square$

Exercise 2. Suppose V is a real vector space with even dimension and $T \in \mathcal{L}(V)$. Suppose $\det T < 0$. Prove that T has at least two distinct eigenvalues.

Proof. From Exercise 1, we know that T has at least one eigenvalue since if it had none the determinant would be positive. Since complex eigenvalues of T_C come in pairs, the sum of the multiplicities of the eigenvalues unique to T_C will be even. Since the sum of the multiplicities of all the eigenvalues of T_C is $\dim V$, which is even, this means that the sum of the multiplicities which are shared between T_C and T , namely the real eigenvalues, is even. Note that 0 is not an eigenvalue of T . Also, by the proof of Exercise 1, we know that the product of the "complex" eigenvalues of T_C is positive, so the product of the real eigenvalues must be negative.

Assume that λ , our one known eigenvalue of T , is in fact the only eigenvalue of T . Then $n = \dim G(\lambda, T_C)$ is even as noted above, which would mean that $\det T = \lambda^n p$ where p is the product of the complex eigenvalues of T_C (repeated according to their multiplicity), and is thus positive. Of course, since λ is real, λ^n is positive if n is even, which would imply that $\det T > 0$, contradicting the hypothesis that $\det T < 0$. Thus, there are at least two distinct eigenvalues of T . $\square$

Exercise 3. Suppose $T \in \mathcal{L}(V)$ and $n = \dim V > 2$. Let $\lambda_1, \dots, \lambda_n$ denote the eigenvalues of T (or of T_C if V is a real vector space), repeated according to multiplicity.

(a) Find a formula for the coefficient of z^{n-2} in the characteristic polynomial of T in terms of $\lambda_1, \dots, \lambda_n$.

(b) Find a formula for the coefficient of z in the characteristic polynomial of T in terms of $\lambda_1, \dots, \lambda_n$.

Proof.

(a) If $n = 3$, then we have

$$\begin{aligned} (z - \lambda_1)(z - \lambda_2)(z - \lambda_3) &= (z - \lambda_1)(z^2 - \lambda_2 z - \lambda_3 z + \lambda_2 \lambda_3) \\ &= z^3 - \lambda_2 z^2 - \lambda_3 z^2 + \lambda_2 \lambda_3 z - \lambda_1 z^2 + \\ &\quad \lambda_1 \lambda_2 z + \lambda_1 \lambda_3 z - \lambda_1 \lambda_2 \lambda_3 \\ &= z^3 - (\lambda_1 + \lambda_2 + \lambda_3)z^2 + \\ &\quad (\lambda_1 \lambda_2 + \lambda_2 \lambda_3 + \lambda_1 \lambda_3)z - \lambda_1 \lambda_2 \lambda_3. \end{aligned}$$

We'll use this as a base case for induction in the proofs of both (a) and (b). Define $p_m = (z - \lambda_1) \cdots (z - \lambda_m)$ for $m \in \{1, \dots, n\}$, and define $c_{m,\ell}$ to be the coefficient of z^ℓ in p_m . We also define $C_m = \{\{j, k\} \mid j, k \in \{1, \dots, m\}, j \neq k\}$. We propose that the coefficient of z^{n-2} is of the form

$$\sum_{\{j,k\} \in C_n} \lambda_j \lambda_k.$$

To prove this, first observe from the above proposed base case that the hypothesis does in fact hold for $n = 3$, so we do in fact have a base case. Now assume that $n > 3$ and the hypothesis holds for all values less than n . Then we can write

$$p_n = (z - \lambda_n)p_{n-1}, \tag{1}$$

so long as each λ_j is indexed appropriately. Observe by (1) that $c_{n,n-2}$ will be precisely $c_{n-1,n-3} - \lambda_n c_{n-1,n-2}$. We know that $c_{n-1,n-2} = -\sum_{j=1}^{n-1} \lambda_j$, and we also know that the hypothesis holds for $c_{n-1,n-3}$. Thus

$$c_{n,n-2} = \left(\sum_{\{j,k\} \in C_{n-1}} \lambda_j \lambda_k \right) + \left(\sum_{\ell=1}^{n-1} \lambda_n \lambda_\ell \right). \tag{2}$$

Since the first summand of (2) contains all necessary combinations of $\lambda_1, \dots, \lambda_{n-1}$ and the second contains all combinations of λ_n with $\lambda_1, \dots, \lambda_{n-1}$, we can see that in fact (2) is

$$\sum_{\{j,k\} \in C_n} \lambda_j \lambda_k$$

as desired.

(b) If $n = 4$, we can take this and get

$$\begin{aligned}
(z - \lambda_1)(z - \lambda_2)(z - \lambda_3)(z - \lambda_4) &= (z - \lambda_4)(z^3 - (\lambda_1 + \lambda_2 + \lambda_3)z^2 + \\
&\quad (\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_1\lambda_3)z - \lambda_1\lambda_2\lambda_3) \\
&= z^4 - (\lambda_1 + \lambda_2 + \lambda_3)z^3 + \\
&\quad (\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_1\lambda_3)z^2 - \lambda_1\lambda_2\lambda_3z - \\
&\quad \lambda_4z^3 + (\lambda_1\lambda_4 + \lambda_2\lambda_4 + \lambda_3\lambda_4)z^2 - \\
&\quad (\lambda_1\lambda_2\lambda_4 + \lambda_2\lambda_3\lambda_4 + \lambda_1\lambda_3\lambda_4)z + \\
&\quad \lambda_1\lambda_2\lambda_3\lambda_4 \\
&= z^4 - (\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4)z^3 + \\
&\quad (\lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_1\lambda_3 + \\
&\quad \lambda_1\lambda_4 + \lambda_2\lambda_4 + \lambda_3\lambda_4)z^2 - \\
&\quad (\lambda_1\lambda_2\lambda_4 + \lambda_2\lambda_3\lambda_4 + \lambda_1\lambda_3\lambda_4)z + \\
&\quad \lambda_1\lambda_2\lambda_3\lambda_4
\end{aligned}$$

Do a similar god-forsaken combinatorial procedure as that in the proof of (a) and you will see (probably) that $c_{n,1}$ is the sum of all three-item combinations of $\lambda_1, \dots, \lambda_n$. $\square$

Exercise 4. Suppose $T \in \mathcal{L}(V)$ and $c \in \mathbf{F}$. Prove that $\det(cT) = c^{\dim V} \det T$.

Proof. Since $\det T = \det \mathcal{M}(T)$, it will suffice to consider the determinant of $\mathcal{M}(cT) = c\mathcal{M}(T)$. Note that $c\mathcal{M}(T)$ multiplies every column of $\mathcal{M}(T)$ by c . Applying 10.39 successively to each column reveals that

$$\det(c\mathcal{M}(T)) = c^{\dim V} \det \mathcal{M}(T)$$

since there are $\dim V$ columns of $\mathcal{M}(T)$. $\square$

Exercise 5. Prove or give a counterexample: if $S, T \in \mathcal{L}(V)$, then $\det(S+T) = \det S + \det T$.

Proof. Consider $S = T = I$. Then

$$\begin{aligned}
\det(S + T) &= \det(\mathcal{M}(S + T)) \\
&= \det(\mathcal{M}(S) + \mathcal{M}(T)) \\
&= \det\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) \\
&= \det\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \\
&= 4
\end{aligned}$$

Since $\det S = \det T = 1$, $\det S + \det T = 2$. Thus

$$\det(S + T) = 4 \neq 2 = \det S + \det T.$$

$\square$

Exercise 6. Suppose A is a block upper-triangular matrix

$$A = \begin{pmatrix} A_1 & & * \\ & \ddots & \\ 0 & & A_m \end{pmatrix},$$

where each A_j along the diagonal is a square matrix. Prove that

$$\det A = (\det A_1) \cdots (\det A_m).$$

Exercise 7. Suppose A is an n -by- n matrix with real entries. Let $S \in \mathcal{L}(\mathbb{C}^n)$ denote the operator on $\mathbb{C}^n$ whose matrix equals A , and let $T \in \mathcal{L}(\mathbb{R}^n)$ denote the operator on $\mathbb{R}^n$ whose matrix equals A . Prove that $\text{trace } S = \text{trace } T$ and $\det S = \det T$.

Proof. Both statements follow trivially from the fact that $\mathcal{M}(T) = \mathcal{M}(T_C)$. $\square$

Exercise 8. Suppose V is an inner product space and $T \in \mathcal{L}(V)$. Prove that

$$\det T^* = \overline{\det T}.$$

Use this to prove that $|\det T| = \det \sqrt{T^*T}$, giving a different proof than was given in 10.47.

Proof. Consider a basis of V as in 8.29. Let $n = \dim V$ and $\lambda_1, \dots, \lambda_n$ the entries along the diagonal of the matrix of T with respect to our chosen basis, i.e. the eigenvalues of T repeated according to their multiplicity. Then $\det T = \det \mathcal{M}(T) = \lambda_1 \cdots \lambda_n$. Note that $\mathcal{M}(T^*)$ is the conjugate transpose of $\mathcal{M}(T)$, so the diagonal of $\mathcal{M}(T^*)$ is $\overline{\lambda_1}, \dots, \overline{\lambda_n}$. Since $\mathcal{M}(T^*)$ is "lower-triangular", we can easily see that the only term which is nonzero in the definition of the determinant is the product of the entries along the diagonal. Thus

$$\det T^* = \overline{\lambda_1} \cdots \overline{\lambda_n} = \overline{\lambda_1 \cdots \lambda_n} = \overline{\det T}.$$

For the second part of the exercise, we proceed with

$$\begin{aligned} |\det T| &= \sqrt{\det T^* \det T} \\ &= \sqrt{\det T^* \det T} \\ &= \sqrt{\det T^* T}. \end{aligned} \tag{1}$$

This gets us close, but not all the way there. Now we need to show that $\sqrt{\det T^* T} = \det \sqrt{T^* T}$.

Redefine $\lambda_1, \dots, \lambda_n$ to be the eigenvalues of $T^* T$, repeated according to multiplicity. Note that the eigenvalues of $\sqrt{T^* T}$ (repeated according to multiplicity) are $\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}$. So

$$\begin{aligned} \sqrt{\det T^* T} &= \sqrt{\lambda_1 \cdots \lambda_n} \\ &= \sqrt{\lambda_1} \cdots \sqrt{\lambda_n} \\ &= \det \sqrt{T^* T} \end{aligned}$$

as desired.

□